medium, the intensity of light coming out (I) decreases exponentially, with increase in the concentration and pathlength of the medium through which it is passed. The fraction of such incident light absorbed by a given solution at a particular wave length is related to the thickness of the absorbing layer i.e., the path length and concentration of the absorbing species. This relation may be combined using Beer-Lambert's law to determine the concentration of analyte in a solution using spectrophotometry.

logIo/I = εcl, where Io and I are the intensities of incident and transmitted light, respectively. l, is the path length, c is the concentration of the absorbing species and ε is known as the molar extinction co-efficient. The expression logIo/I is absorbance.

10.6.2 Photo electric colorimeter

Principle:A beam of light with a precise wavelength is passed through the sample through a mono-chromator and lens, which navigate the coloured light to the output device which measures the extent of colour as compared to a standard. A microprocessor then calculates the absorbance or percent transmittance. If the concentration of the solution is greater, more light will be absorbed, which can be identified by measuring the difference between the amount of light at its origin and that after passing through the solution.

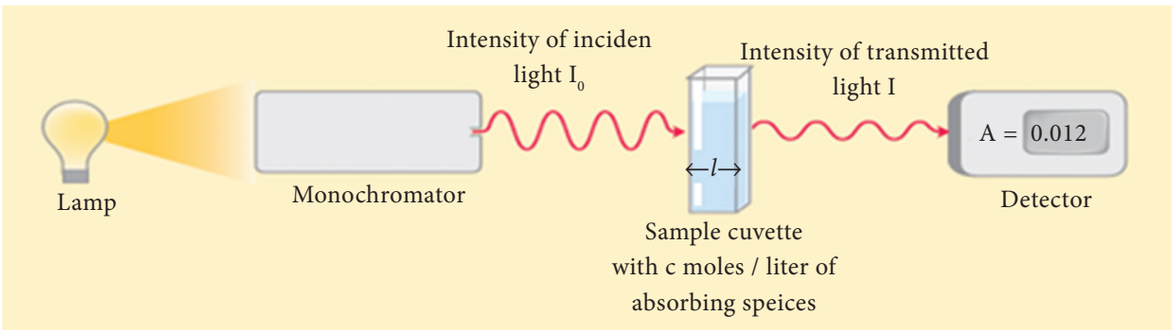


Figure 10.15 Spectrophotometry

Components of a Colorimeter :

A schematic representation of the working of the colorimeter is shown in Fig.10.16

The essential parts of a colorimeter are: a) light source, which is from a tungsten lamp and the light passes through a slit b) an aperture which can be adjusted accordingly c) a filter - where the light from the lamp source converts polychromatic light to a monochromatic light d). photocell or tube which converts the light energy to electrical energy e) a detector measures the light which was transmitted by the sample f) cuvette – a glass tube to place the solution.

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